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REVIEW

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Deep generative molecular design and its value in modern drug discovery

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ABSTRACT

Introduction: Deep generative models are reshaping *de novo* drug design by enabling creation of novel, property-optimized molecules beyond traditional chemical libraries. Advances in deep learning, molecular representation learning, and structure-aware modeling now enable algorithms to propose molecules that satisfy complex pharmacological constraints, accelerating hit identification.

Areas covered: This review outlines recent advances in generative molecular design, including neural network-based frameworks, reinforcement learning systems, diffusion models, and language model-based transformers. The authors outline how each class generates and optimizes molecular structures and review generative AI's practical applications in drug discovery, illustrating translational progress. Current bottlenecks are critically analyzed alongside emerging solutions. This review is based on a systematic literature search conducted in Google Scholar and PubMed, covering studies published up to December 2025.

Expert opinion: Generative AI's greatest promise lies not in generating more molecules, but in generating better hypotheses, structures that are synthetically accessible, biologically plausible, optimized across potency, selectivity, and pharmacokinetics. The next phase will be led by multimodal foundation models capable of reasoning jointly about chemistry, protein structure, and cellular response, supported by automated synthesis and high-throughput experimentation. As these components are integrated, generative molecular design will guide lead optimization and reshape how new therapies are discovered and developed.

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ADMET optimization; *de novo* drug design; diffusion models; generative AI; molecular generation; structure-based drug discovery

1. Introduction

Drug discovery is undergoing a fundamental transformation as AI moves from predictive modeling to generative design [1]. For decades, computational drug discovery has focused on predicting molecular properties and performing virtual screening within predefined chemical libraries, approaches that refine existing knowledge rather than generate new molecular possibilities [2]. Traditional structure- and ligand-based design methods rely on iterative screening within a relatively narrow region of chemical space, enabling optimization of known scaffolds but limiting efficiency and the discovery of fundamentally new molecules [3,4]. The emerging capacity to explore the vast and sparsely sampled chemical landscape, estimated to contain over 10^{60} potential compounds, marks a paradigm shift in how hit and lead candidates are conceived [5]. In contrast, deep generative molecular design focuses on creating new molecules rather than screening existing ones. By learning the statistical and physicochemical principles of molecular space, these models can generate novel, synthesizable compounds with properties optimized for therapeutic performance [1,6–8].

At the core of this transformation are deep learning (DL) models that learn probability distributions over chemical representations rather than relying on manually engineered descriptors or fingerprints. These generative models learn internal latent spaces that capture the 'grammar' and physicochemical relationships underlying molecular structure [9]. Molecules may be encoded as sequences (e.g. SMILES), as graphs where atoms are nodes and bonds are edges, or as 3D coordinate sets that reflect spatial geometry. Sequence-based models borrow architectures from natural language processing, such as transformers and recurrent networks, to interpret molecular syntax as a chemical language [10]. Graph-based models, rooted in geometric DL, represent atoms as nodes and bonds as edges, capturing topological and electronic context more explicitly [11,12]. More recently, 3D-aware models have incorporated spatial coordinates to learn atomic interactions and conformational flexibility, bridging the gap between ligand design and protein-ligand binding physics [13–16]. By capturing both structural symmetry and quantum-level interactions, these approaches outperform earlier

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Article highlights

- Deep generative models are redefining *de novo* molecular design by autonomously creating chemically valid and pharmacologically relevant compounds.
- Neural network (NN)-based generative frameworks and reinforcement learning (RL)-guided graph models capture nonlinear structure-activity relationships (SARs) and enable multi-objective molecular optimization beyond what rule-based approaches allow.
- Diffusion-based generative models introduce physically grounded 3D structure generation, supporting *de novo* scaffold synthesis, conformational exploration, and pocket-aware ligand design.
- Language model (LM)-based approaches provide scalable, flexible control over chemical syntax and can generalize across large chemical datasets to enable rapid hypothesis generation and fine-tuned property optimization.
- Despite strong *in silico* performance, a translational gap remains.
- Closing this gap will require tighter coupling of generative modeling with automated synthesis, high-throughput biological testing, and iterative model refinement to build closed-loop discovery pipelines.

variational autoencoders (VAE) and generative adversarial networks (GAN) in producing stable, drug-like conformations [7,17,18].

The data required by these models vary widely, from curated molecular libraries and binding pocket geometries to omics-derived target features. Their resulting applications can range from complete molecular structures to optimized analogs satisfying predefined constraints such as solubility, target affinity, or synthetic accessibility [19]. Many frameworks incorporate RL or conditional training to steer generation toward molecules with specific therapeutic or Absorption, Distribution, Metabolism, Excretion, and Toxicity (ADMET) profiles, effectively uniting generative creativity with goal-directed optimization [20,21]. Some approaches integrate multimodal data, linking chemical and biological representations (e.g. gene expression signatures, protein embeddings) to design compounds predicted to reverse disease phenotypes. These hybrid models move beyond traditional quantitative SAR (QSAR) or docking workflows toward a comprehensive modeling of drug-target-phenotype interactions [22,23].

Encouragingly, the field is transitioning from computational novelty to translational success [24,25]. Several AI-designed and AI-optimized molecules have already entered clinical trial phases, marking a significant milestone for data-driven drug discovery [24–27]. These developments underscore the growing potential of generative AI to accelerate the discovery pipeline, from target identification to lead optimization, while reducing dependence on traditional trial-and-error screening. Recent analyses indicate that AI-derived molecules show promising clinical performance, with Phase I success rates reaching 80–90% in selected disclosed programs, substantially higher than historical averages [28]. This improvement may stem from AI models' ability to design drug-like molecules with optimized ADMET properties, reducing early-stage attrition. Although Phase II success rates (~40%) align with industry norms, most discontinuations appear driven by business or funding constraints rather than efficacy or safety failures, underscoring the robustness of AI-generated leads and hinting at their growing translational potential [28,29]. Despite their promise, deep generative models face several conceptual and practical limitations. A measurable translation gap still remains between *in silico* predictions and consistent experimental validation, reflecting the inherent complexity of biological systems. Ongoing progress in coupling AI-driven design with empirical testing promises to narrow this gap, ushering in an era of adaptive, data-rich discovery pipelines capable of delivering therapeutics with unprecedented speed, precision, and biological fidelity [30].

In this review, we first discuss the core generative molecular design approaches, including NN-based, diffusion-based, and LM-based frameworks, emphasizing how these paradigms differ from traditional rule-based or descriptor-driven methods. We then examine the applications of generative AI in modern drug discovery, highlighting its capabilities in generating novel molecular scaffolds, exploring new conformations, and optimizing key physicochemical and pharmacological properties. These are supported by case studies demonstrating their integration into discovery pipelines. Finally, we discuss the limitations and current bottlenecks that hinder broader translation. Together, these perspectives provide a comprehensive overview of how cross-domain AI innovations are reshaping molecular design, while underscoring the need for continued improvement to ensure reliable and experimentally grounded progress in drug discovery.

This review is based on a structured but non-systematic survey of the scientific literature conducted using Google Scholar and PubMed, covering publications available up to December 2025. The following keyword combinations were used: 'generative AI drug discovery,' 'DL drug discovery,' 'AI-designed molecule,' '*de novo* molecular generation,' 'diffusion models drug discovery,' 'reinforcement learning drug design,' 'transformer molecular generation,' 'structure-based generative design,' 'AI clinical trial drug discovery,' 'AI-designed compound,' and related variants. To identify translational case studies, targeted searches were additionally performed using company-specific terms including 'Insilico Medicine,' 'BenevolentAI,' 'Exscientia,' 'Schrödinger,' 'Recursion Pharmaceuticals,' 'Atomwise,' and 'Generate Biomedicines' in combination with 'clinical trial,' 'Phase I,' 'Phase II,' 'AI-designed drug,' and 'generative model.' We prioritized peer-reviewed publications and regulatory or clinical announcements documenting experimental validation or clinical progression of AI-designed compounds. Together, these sources provide a representative and up-to-date perspective on both methodological developments and translational applications of generative molecular design.

2. Generative molecular design approaches

Generative molecular design has emerged as a crucial catalyst in modern drug discovery, offering a transformative route to accelerate the identification of therapeutically viable compounds. Its importance lies in uncovering entirely novel chemical and structural solutions tailored to specific biological and physicochemical objectives while minimizing experimental and financial costs. Within this framework, several methodological paradigms have taken shape, each contributing

unique capabilities to the field. VAE and GAN methods capture complex nonlinear SARs critical for optimization [31,32]; diffusion-based approaches excel at learning physically consistent molecular transformations and generating realistic 3D conformations [33–35]; and LM-based techniques leverage the expressive power of molecular syntax to produce syntactically valid and chemically diverse compounds [5,36,37]. While each approach demonstrates distinct strengths, none is without limitations. Understanding these complementary strengths and shortcomings is essential for building the next generation of generative frameworks capable of translating computational design into real-world therapeutic success.

2.1. NN-based approaches: data-driven molecular generation through learned chemical representations and adaptive optimization

NN-based generative molecular design encompasses a diverse family of modeling approaches that have collectively transformed how chemical space is explored and optimized. The Junction Tree VAE (JT-VAE) remains a foundational model that builds molecules hierarchically, first generating a tree of valid chemical substructures and then assembling them into full molecular graphs [31]. This two-stage process enforces chemical validity during decoding and provides a continuous latent space for optimizing molecular properties, making it particularly effective for scaffold-controlled generation and fine-tuning of lead compounds. GraphVAE offers a more direct, nonautoregressive approach by learning probabilistic adjacency tensors that define atom and bond configurations [38]. It performs well for small molecules and conceptually demonstrates the feasibility of graph-based generative modeling, although it scales poorly to larger systems due to graph-matching overhead.

MolGAN extends GANs to molecular graphs and incorporates RL-based rewards to guide the generator toward compounds with desirable chemical or pharmacological properties [32]. It achieves high validity and uniqueness on small datasets such as QM9 but can exhibit instability during training and limited structural diversity. The Graph Convolutional Policy Network (GCPN) adopts an RL framework where molecules are built step by step through atom and bond additions under chemical valence rules [39]. This allows explicit control over synthesis-like construction and multi-objective optimization, though it requires extensive training and rollout sampling, which can limit efficiency. Another important line of work, represented by GraphAF, employs normalizing flows to map molecular graphs through invertible transformations, providing exact likelihoods and enabling gradient-based optimization [40]. This approach maintains strong chemical validity and supports property optimization through RL fine-tuning but is computationally demanding and depends on large, high-quality datasets.

Together, these NN-based methods represent complementary strategies for molecular design. JT-VAE and GraphVAE provide smooth latent spaces for optimization and constrained reconstruction [31,38], MolGAN and GCPN emphasize generative flexibility and goal-directed synthesis [32,39], and flow-based approaches like GraphAF balance probabilistic rigor with controllability [40]. Each has distinct advantages; VAEs offer interpretability and optimization potential [31,38], GANs and reinforcement-based models excel in generating property-optimized structures, and flow models bring stability and tractable likelihood estimation [32].

However, all share common limitations, including data dependence, limited 3D awareness, and challenges in linking generated molecules to realistic synthesis or binding contexts. Combining these architectures with physics-informed constraints, 3D modeling modules, and experimental feedback remains an active area of research aimed at bridging the gap between computational creativity and practical drug discovery.

2.2. Diffusion-based approaches: physically consistent 3D generation and exploration of complex molecular conformational space

Diffusion-based generative modeling has rapidly become a cornerstone of modern molecular design, introducing probabilistic frameworks that learn to iteratively denoise molecular structures toward physically meaningful configurations.

A defining advance of modern diffusion models is the incorporation of SE(3)- and E(3)-equivariant neural architectures, which explicitly enforce rotational and translational symmetries inherent to molecular systems. Because molecular energies and interactions are invariant under global rotations and translations, generative models that violate these symmetries can produce distorted or physically inconsistent conformations. Equivariant diffusion models preserve these symmetries throughout the denoising process, ensuring that predicted atomic updates transform consistently under coordinate changes. This architectural constraint improves geometric fidelity, conformer realism, and docking pose accuracy by preventing the generation of nonphysical geometries and stabilizing learned interaction patterns. In practice, equivariant frameworks such as EDM [33], DiffDock [41], DiffLinker [42], and DiffSBDD [43] have demonstrated superior performance in 3D molecular generation, fragment linking, and pocket-conditioned ligand design, producing structures that better reproduce experimental binding modes and conformational ensembles.

Among the most established models, the E (3)-Equivariant Diffusion Model (EDM) enables end-to-end 3D molecular generation by learning to jointly denoise atom types and coordinates with full rotational and translational equivariance, producing realistic 3D molecules rather than SMILES-based strings (Figure 1) [33]. GeoDiff focuses on conformational generation for known molecular graphs, efficiently sampling stable conformers useful for docking and molecular dynamics (MD) initialization [34]. DiffLinker, on the other hand, extends diffusion modeling to fragment-based drug design, learning to connect multiple fragments in 3D without predefined attachment points and supporting pocket-conditioned design for structure-based applications [42]. Building on this framework, DiffSBDD advances the concept further by directly generating full ligand structures within protein binding pockets [43]. It integrates an equivariant diffusion model with spatial conditioning on protein atomic coordinates, allowing the network

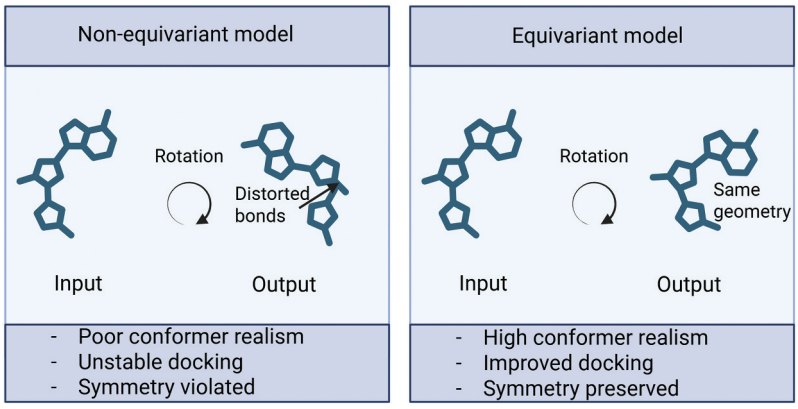


Figure 1. Effect of equivariance in diffusion-based molecular generation.

Illustration of how SE(3)/E(3)-equivariant architectures preserve rotational and translational symmetries during molecular generation. In non-equivariant models (left), applying a global rotation or translation to the input molecule can lead to inconsistent and distorted output geometries. In contrast, equivariant diffusion models (right) ensure that denoising updates transform consistently under coordinate changes, yielding identical output geometries that respect molecular symmetries. This property improves geometric fidelity, conformer realism, and docking pose stability, producing molecular structures that better reproduce experimentally observed binding modes. Created in BioRender. Ozdemir, S. (2026) https://BioRender.com/lo6mfgm.

to learn realistic protein-ligand interaction patterns such as hydrogen bonding and hydrophobic packing. Complementing the aforementioned methods, DiGress performs discrete diffusion directly in graph space, modeling the addition, removal, and transformation of atoms and bonds, ideal for constructing chemically valid graphs before introducing 3D structure [44]. Finally, Jing et al. introduces torsional diffusion, a physically intuitive approach by operating on torsion-angle manifolds, generating conformers through diffusion over bond rotations while maintaining molecular connectivity, offering high accuracy and efficiency in exploring conformational flexibility [35].

The rapid adoption of predicted protein structures, driven by advances such as AlphaFold and ESMFold, has substantially expanded the scope of structure-based generative ligand design, enabling modeling for targets lacking experimental structures [45–48]. However, the reliability of ligand generation conditioned on predicted structures remains an important consideration. While predicted models often achieve high global fold accuracy, local errors in side-chain positioning, loop regions, and binding-site geometry can materially affect ligand pose prediction and interaction modeling [45,48]. Consequently, generative models conditioned on predicted structures may propagate these uncertainties into proposed ligand conformations. Emerging best practices therefore emphasize the use of structural ensembles, uncertainty-aware conditioning, and experimental refinement, when possible, particularly for high-stakes lead optimization [49–51]. As structure prediction tools continue to improve, their integration with generative design will remain powerful, but careful validation against experimental data is essential for ensuring translational reliability.

Across these approaches, applications range from full 3D molecular generation [33] to conformer sampling [34,35], fragment linking [42], and discrete graph-level de novo design [44]. EDM and GeoDiff excel in geometric fidelity through E(3)-equivariance, ensuring that generated molecules respect spatial symmetries [33,34]. DiffLinker and DiffSBDD uniquely enables structural assembly within protein pockets, bridging generative chemistry and structure-based design [42,43], while DiGress offers greater control over topological editing and graph exploration [44]. Torsional diffusion stands out for its ability to efficiently explore the conformational landscape without requiring explicit force-field integration [35].

Despite these advances, common limitations persist. Models like EDM and DiffLinker rely heavily on large, high-quality 3D datasets [33,43]; GeoDiff and torsional diffusion require downstream physics-based validation to assess binding or synthetic feasibility [34]; and DiGress, while versatile in graph generation, must be coupled with 3D reconstruction for realistic molecular geometries [44].

2.3. LM-based approaches: sequence-driven molecular design through contextual learning and scalable generative control

LM-based generative design treats molecules as sequences (typically SMILES or SELFIES) and learns a probability distribution over tokens for *de novo* design and goal-directed optimization.

RNN-LSTM by Gupta et al. presents an early recurrent NN (RNN) models using a long short-term memory (LSTM) trained to generate valid SMILES and then steered by property predictors or RL to propose drug-like libraries [36]. This idea matured into the REINVENT family, which couples a generative language model with reinforcement learning to bias sampling toward multi-objective molecular profiles. Early generations (e.g. REINVENT 2.0) used recurrent architectures, while later iterations introduced transformer-based models to improve long-range token dependencies and controllability [52]. REINVENT 2.0 marked a major stabilization point in the framework: it provided a modular, production-ready implementation with exploration/exploitation modes, a dynamic multi-property scoring function, diversity filters to mitigate mode collapse, and RL loops for adaptive sampling. These design principles formed the foundation for subsequent transformer-based upgrades (e.g. REINVENT-Transformer) and later major

releases such as REINVENT 4, which extend the platform with improved training pipelines, scalability, and compatibility with modern predictive and synthesis-planning tools [37,53]. Across versions, REINVENT remains one of the most widely used open-source platforms for de novo drug design, valued for its flexibility across target classes and optimization objectives.

Pure transformer decoders trained on next-token prediction further improved sample quality and conditional generation, demonstrating strong validity and novelty on drug-like molecular datasets while enabling prompt-style conditioning for properties or substructures [54]. At larger scale, sequence-to-sequence models such as MegaMolBART leverage BART pretraining on massive SMILES datasets to learn rich latent embeddings that support both generation and downstream tasks; these models are now standard components in industrial pipelines [55]. Newer GPT-style approaches (i.e. MolGPT) push autoregressive generation with task prompts or property-aware fine-tuning to target specific chemical regions or optimization goals [5,56]. Robust string representations like SELFIES are often paired with these LMs to eliminate invalid outputs and stabilize training, making token-based generation far more reliable at scale [57,58].

Compared head-to-head, RNN-LSTM and early REINVENT models are simple, fast, and easy to fine-tune for single- or multi-objective optimization, but they can struggle with long-range grammar and are sensitive to string fragility without SELFIES [36,37]. Transformer decoders and REINVENT-Transformer generally yield higher validity and better controllability via attention and flexible conditioning, yet they require larger datasets and compute, and still operate mostly in 1D token space without explicit 3D awareness [37,52,54]. MegaMolBART-style models excel at representation learning and can be adapted to many tasks, but inference cost is higher and chemical realism still depends on downstream filters and structure-based checks [55]. GPT-style models such as MolGPT are strong for promptable, property-aware generation and few-shot adaptation, though they can overfit rewards or drift toward unrealistic chemotypes without synthesizability constraints and pocket/context conditioning [5].

Across all LM-based approaches, common limitations are lack of inherent 3D or protein-context grounding, dependence on data quality, and the need for retrosynthesis, docking/MD, and experimental feedback to translate text-based creativity into viable leads. Table 1 summarizes the major NN-, diffusion-, and LM-based generative molecular design frameworks, highlighting their core methodologies, primary applications, and key limitations.

2.4. Traditional vs generative approaches in molecular design

Traditional computational methods, such as molecular docking, QSAR, pharmacophore modeling, and virtual screening, have long supported drug discovery by translating chemical intuition into rule-based, descriptor-driven models. These approaches enabled systematic exploration of chemical space but rely on predefined features and empirical scoring functions. As a result, they often optimize one property at a time, require repeated design-test cycles, and struggle with multi-objective trade-offs such as potency, selectivity, and pharmacokinetics. Moreover, their dependence on human-crafted descriptors limits their ability to capture the complex, nonlinear relationships that govern molecular structure and biological response. Despite these limitations, they laid the foundational concepts that generative approaches now build upon.

Generative AI-based molecular design represents a shift from hypothesis-driven to data-driven exploration. Rather than evaluating predefined libraries, generative models learn

Table 1. Overview of generative molecular design approaches, their core methods, primary functions, and limitations.

Category	Name	Function	Limitations
NN-based	JT-VAE [31]	Scaffold-controlled molecule generation and optimization	Constrained to predefined fragment vocabularies; limited scalability
	GraphVAE [38]	Direct graph generation for small molecules	Poor scalability; graph-matching overhead
	MolGAN [32]	Property-optimized molecular generation	Unstable training; limited diversity and size
	GCPN [54]	Stepwise synthesis-like molecule construction	Computationally expensive; requires extensive rollout sampling
	GraphAF [40]	Likelihood-based molecular generation with property optimization	Computationally demanding; dataset dependent
Diffusion-based	EDM [33]	End-to-end 3D molecular generation with rotational and translational invariance	Requires large, high-quality 3D datasets; lacks synthesis feasibility checks
	GeoDiff [34]	Efficient conformer generation for docking and MD initialization	Limited to known graphs; needs physics-based validation
	DiffLinker [42]	Fragment-based drug design and pocket-conditioned assembly	Dependent on structural data; limited control over global drug-likeness
	DiffSBDD [43]	*De novo* ligand generation within protein binding pockets	Requires accurate protein structures; high computational cost
	DiGress [44]	Graph-level *de novo* molecular design	Needs subsequent 3D reconstruction for geometry realism
	Torsional diffusion [35]	Conformer generation capturing molecular flexibility	Limited by bond-rotation data; lacks full energy landscape modeling
LM-based	RNN-LSTM [36]	*De novo* molecular generation with RL-based optimization	Limited long-range dependency modeling; string fragility
	REINVENT family [37,52,53]	Multi-objective molecular optimization with exploration/exploitation control	Requires large datasets; operates in 1D token space without 3D context
	MegaMolBART [55]	Representation learning and generative fine-tuning	Computationally intensive; chemical realism depends on downstream checks
	MolGPT [5]	Promptable, property-aware molecular generation	May overfit rewards; lacks synthesizability and 3D context
	SELFIES-integrated models [57]	Valid sequence generation eliminating invalid SMILES	Still text-based; requires structure-based refinement

chemical and structural relationships directly from large datasets, enabling the proposal of novel compounds that extend beyond known scaffolds. NN-based models, including VAEs, GANs, and reinforcement learning systems, support scaffold-controlled design, fragment linking, and multi-parameter optimization. Diffusion-based frameworks generate physically plausible 3D structures and capture spatial interaction patterns, though they still require physics-based refinement. LM approaches treat molecules as sequences, allowing scalable generation and flexible conditioning, but they lack inherent 3D or protein-context awareness and therefore benefit from integration with structure-based methods. Together, these generative strategies expand chemical exploration while aiming to improve efficiency and design quality beyond traditional methods.

Both traditional and generative approaches continue to play important roles in drug discovery, however, their practical capabilities differ. To clarify these complementary roles, Table 2 provides a structured comparison across key methodological dimensions relevant to molecular design, including sampling efficiency, physical grounding, scalability, and application scope, with representative real-world examples. Rather than positioning generative AI and traditional methods as opposing approaches, this comparison highlights how these tools occupy different roles within modern discovery pipelines. In practice, generative frameworks frequently incorporate components derived from traditional computational chemistry, such as docking, QSAR models, scoring functions, and MD-based refinement, forming integrated, multi-step workflows rather than mutually exclusive paradigms. Across model classes, generative molecular design methods can be understood along a set of shared conceptual dimensions that govern their practical utility: representation fidelity, controllability, physical grounding, and optimization robustness. Sequence- and graph-based models offer high scalability and controllability but limited physical realism; diffusion and equivariant models provide superior geometric fidelity and physical consistency at greater computational cost; RL frameworks enable explicit objective-driven optimization but remain sensitive to reward design and stability; and physics-based simulations deliver rigorous thermodynamic grounding but suffer from limited sampling efficiency and scalability. The choice of model class therefore depends strongly on the discovery context: early ideation favors scalable generative models for chemical space exploration, while structure-based lead optimization benefits from diffusion and equivariant frameworks, and late-stage refinement requires integration with physics-based methods and synthesis-aware constraints. Framing generative methods within these shared dimensions provides a unified perspective for selecting appropriate tools under specific drug discovery scenarios.

3. Application of generative AI in modern drug discovery

Drug discovery demands the ability to identify effective molecules within an extraordinarily vast and sparsely sampled chemical universe [63,64]. Traditional discovery pipelines can only probe a negligible portion of this space and often fail to capture the complex, nonlinear relationships between

Table 2. Roles of generative AI models, physics-based MD simulations and integrated generative-physics workflows in molecular design.

Dimension	Generative AI models	Conventional MD simulations	Integrated generative-physics workflows	Representative real-world examples
Primary objective	*De novo* molecular generation, optimization, and hypothesis exploration	Physics-based simulation of molecular motion and energetics	End-to-end discovery workflows	DiffDock [41], GeoDiff [34], AMBER [59], GROMACS [60]
Sampling efficiency	Extremely high; can generate 10^4 –10^6 candidate structures per day	Computationally expensive; microsecond-millisecond timescales per system	High	Insilico Medicine; reducing the time from project initiation to a preclinical candidate nomination to ~18 months, compared to the traditional industry average of 2.5–4 years [25].
Exploration of chemical space	Broad, data-driven exploration beyond known scaffolds	Limited to local conformational landscapes of predefined molecules	Broad exploration followed by physics-based narrowing and prioritization	RFdiffusion [61]
Physical grounding	Learned statistical approximations of molecular physics; often combined with post hoc refinement	Explicit force fields and thermodynamic principles	Combines statistical learning with physics-based validation	DiffSBDD [43] + MD refinement
Scalability	Highly parallelizable on GPUs	Limited by system size and simulation length	High	Large-scale generative screening
Conformer diversity	High diversity via stochastic generation	Restricted by energy barriers and simulation time	Generative conformer proposals refined and validated by MD ensemble sampling	Torsional diffusion
Conditioning capability	Can be conditioned on protein pockets, target properties, and constraints	Indirect via biasing potentials and restraints	Multi-level conditioning combining property rewards, structural constraints, and thermodynamic validation	Pocket-conditioned diffusion [62]
Practical role in pipelines	Early-stage ideation and rapid hypothesis generation	Late-stage validation and physical refinement	Iterative design-refine-validate cycles	Generative proposal → MD validation workflow

molecular structure and biological response. Generative AI addresses these limitations by modeling the underlying principles of chemistry, biology, and physics to create rather than merely screen molecules. In this section, we discuss the key applications of generative AI that were once prohibitively difficult, including *de novo* design of novel scaffolds, generation of realistic conformational ensembles, prediction of multiparametric properties such as ADMET, and target-specific ligand optimization as well as recent case studies demonstrating its translational potential. By unifying these tasks into a single, data-driven framework, generative modeling offers a way to accelerate discovery, reduce experimental burden, and uncover chemical innovations that might otherwise remain inaccessible.

3.1. De Novo molecular generation and scaffold design, and translational examples

Designing entirely new compounds that are both synthesizable and biologically active has long been one of the hardest problems in medicinal chemistry [64]. The challenge stems from the astronomical size of chemical space, the sparsity of experimentally validated data, and the need to satisfy multiple often-conflicting objectives such as potency, selectivity, and pharmacokinetic balance [63].

Generative AI has changed this process by enabling the creation of entirely new chemical entities rather than merely evaluating preexisting libraries. Instead of searching within fixed chemical scaffolds, these models learn how to propose structures that satisfy multiple design constraints simultaneously [37,40]. RL and probabilistic generative frameworks allow the exploration of chemical space in a goal-directed manner, biasing molecular design toward desired therapeutic profiles [37]. Structure-conditioned generative models can further incorporate 3D information about protein binding pockets, protein interfaces enabling the synthesis of ligands that adopt plausible binding poses and interaction geometries [42,43]. Meanwhile, large-scale sequence-based generative models trained on vast chemical datasets can rapidly propose thousands to millions of novel molecules and refine them through property-based conditioning or predictive feedback [5,56].

Recent advances in protein-pocket conditioning provide concrete case studies demonstrating that incorporating structural context substantially improves the realism and practical utility of generative molecular models. DiffSBDD's equivariant diffusion framework directly generates ligands within protein binding pockets and showed that pocket-conditioned models produce ligand conformations and interaction patterns that more closely reproduce experimentally observed binding modes than unconditioned generators [43]. DiffLinker, which conditions generation on both molecular fragments and pocket geometry, enables the design of chemically valid linkers that preserve binding geometry within complex protein environments [42]. DiffDock demonstrates that structure-conditioned diffusion models significantly outperform classical docking approaches in predicting accurate ligand poses across diverse protein-ligand complexes [41]. Moreover, Hu et al. show that an equivariant, pocket-conditioned diffusion framework, DiffGui, significantly improves conformer stability, RMSD agreement with experimentally optimized structures, and protein-ligand interaction similarity across the PDBbind benchmark, outperforming multiple state-of-the-art baselines [62]. Together, these studies illustrate how protein-pocket conditioning enhances conformer realism, binding pose accuracy, and translational relevance in real drug-design workflows.

In parallel with scaffold generation, recent generative models construct molecular conformers directly in continuous 3D space rather than relying on conformer enumeration or brute-force molecular dynamics sampling. Diffusion-based approaches such as GeoDiff [34] and equivariant diffusion models [33] initialize molecules as randomized atomic coordinates and iteratively denoise them toward physically plausible geometries, learning statistical constraints on bond lengths, angles, torsions, and nonbonded interactions from large collections of experimentally resolved structures. Complementarily, torsional diffusion models operate directly in torsion-angle space, enabling efficient exploration of conformational flexibility while preserving molecular connectivity. These learned generative processes rapidly produce diverse, low-energy conformers that approximate physically realizable molecular states, improving sampling efficiency and providing high-quality initial structures for downstream docking, refinement, and experimental evaluation. Figure 2 visually summarizes the conformer construction pipeline from noise initialization to physically plausible 3D conformers.

A critical advance in the practical deployment of generative molecular design is the incorporation of retrosynthetic feedback loops that connect molecular generation with chemical feasibility. Modern pipelines increasingly integrate route-aware synthesis planning and feasibility scoring directly into the generation process, ensuring that proposed molecules are

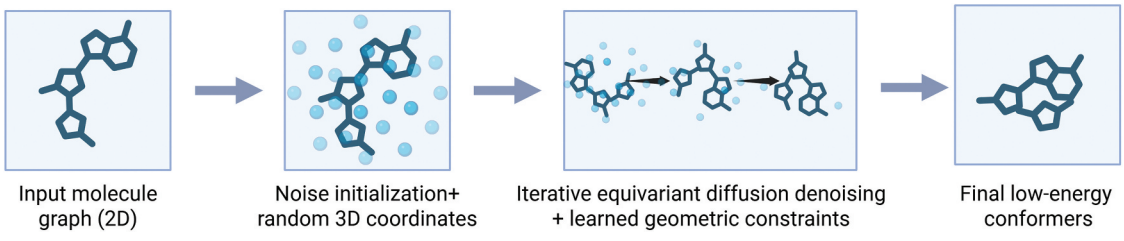


Figure 2. Generative construction of molecular conformers using diffusion-based models.

Starting from an input molecular graph, atoms are embedded into randomized 3D coordinates and iteratively denoised using an equivariant diffusion process that enforces learned geometric constraints, producing physically plausible low-energy conformers for downstream modeling and evaluation. A complementary torsional diffusion process enables efficient exploration of conformational flexibility while preserving molecular connectivity. *Created in BioRender. Ozdemir, S. (2026) https://BioRender.com/lo6mfgm.*

not only biologically promising but also experimentally accessible. Generative models are now coupled with automated retrosynthesis engines and reaction prediction systems to evaluate synthetic routes, rank candidates by synthetic tractability, and penalize designs that require impractical or low-yield transformations. These systems enable iterative design-make-test cycles in which generative models propose candidates, synthesis planning modules assess feasibility, experimental data are collected, and the resulting outcomes are fed back into the model to refine subsequent generations. This closed-loop framework reduces wasted synthesis effort, accelerates lead optimization, and improves the likelihood that computationally designed molecules can advance toward experimental validation and clinical development.

These advances allow researchers to traverse chemical space far more efficiently, uncovering scaffolds unseen in traditional databases. However, the challenge of translating computational novelty into experimentally viable compounds remains. Integrating generative design with retrosynthetic analysis, synthetic accessibility prediction, and automated synthesis planning is now key to ensuring that AI-generated molecules can move from *in silico* ideation to laboratory reality.

One widely discussed demonstration of this approach comes from Insilico Medicine. Their DDR1 kinase inhibitor campaign used generative RL to identify potent chemotypes in less than two months, dramatically outperforming conventional discovery timelines [26]. More recently, Insilico's first-in-class compound INS018_055 advanced to Phase II clinical evaluation for idiopathic pulmonary fibrosis, representing one of the earliest generative-AI-designed molecules to reach this milestone [24,25]. BenevolentAI has similarly applied generative and predictive modeling to produce BEN-2293, a pan-Trk inhibitor now in clinical development [65]. These examples highlight the accelerating translational potential of generative molecular design when coupled with fast experimental iteration.

While these high-profile successes are encouraging, it is important to emphasize that the causal contribution of generative modeling to clinical outcomes cannot be isolated from other decisive factors, including target selection, medicinal chemistry expertise, experimental infrastructure, organizational resources, and business-driven prioritization. Reported success rates largely reflect a small and highly curated subset of programs that have advanced far enough to be disclosed publicly, and thus are likely subject to selection and survivorship bias. Moreover, negative outcomes, failed campaigns, and terminated AI-driven programs are rarely reported in the literature, limiting the ability to rigorously quantify the independent impact of generative AI on clinical success. As such, current evidence should be interpreted as indicative of promise rather than definitive proof of causal advantage.

3.2. Capturing conformational dynamics and binding flexibility, and real-world applications

While this review primarily focuses on molecular (ligand) design, recent advances in protein conformational ensemble generation are discussed here insofar as they directly inform structure-aware generative ligand design and binding-mode prediction. Biomolecular function emerges not from static structures but from the distribution of conformational states that molecules and proteins explore, therefore understanding the dynamic conformational landscape of molecules and proteins is crucial. Binding affinity, allosteric regulation, cryptic pocket formation, induced-fit mechanisms and drug resistance mechanisms are all governed by dynamic structural rearrangements. However, traditional methods such as long-timescale MD simulations are computationally demanding and often capture only a narrow subset of physiologically relevant states.

Generative AI offers an alternative by learning statistical and physical constraints associated with molecular motion. Recent generative models construct conformers by reversing noise applied to atomic coordinates or torsion-angle spaces, capturing realistic molecular flexibility with substantially improved sampling efficiency compared to conventional MD [34,35]. Other extensions incorporate experimental constraints and mutation-dependent perturbations to produce ensembles that reflect slow, biologically relevant conformational transitions [49,50]. These approaches complement physics-based refinement rather than replace it, enabling rapid hypothesis generation about binding-induced structural rearrangements, allosteric shifts, and cryptic pocket formation that are often inaccessible through traditional simulation alone [66].

Several recent studies have shown that generative sampling of receptor flexibility can reveal alternative binding pockets and allosteric sites that are difficult to identify through static crystallography or short MD simulations. For example, structure-conditioned diffusion models have been used to map cryptic and transient binding grooves in kinases and G protein-coupled receptors (GPCRs), enabling the design of ligands that selectively target otherwise hidden conformational states [67,68]. These models capture activation-dependent pocket formation and dynamic side-chain rearrangements that are often missed by static crystal structures. Similarly, generative conformational ensemble approaches have been used to distinguish active versus inactive protein states in signaling proteins, guiding the rational development of state-selective inhibitors that modulate functional conformational populations [69]. Complementing these generative strategies, recent work with DeepAllo demonstrated that protein LM-derived representations, combined with pocket descriptors, can accurately identify allosteric regions even when only predicted structural information is available, achieving strong discrimination of functional pockets without relying solely on static crystallographic annotations [66]. The study further showed that attention patterns learned by the model highlight residues proximal to modulatory sites and can recover cryptic or distal regulatory pockets, strengthening the rationale for using learned conformational context to guide structure-aware ligand design.

3.3. Property and ADMET-aware molecular optimization, and practical demonstrations

Even molecules with strong binding may fail due to poor ADMET properties. Predicting these properties has historically been challenging because they depend on complex, nonlinear

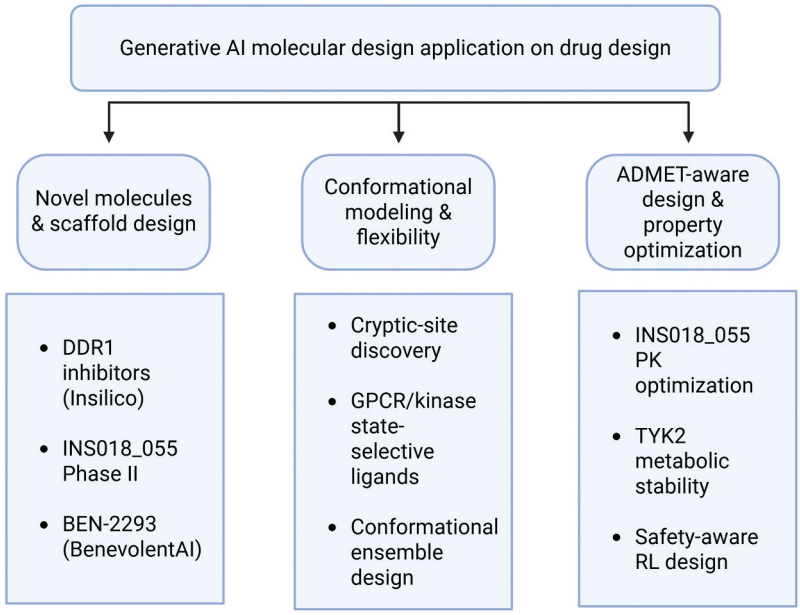


Figure 3. The major application domains of generative AI in drug design.

Novel molecule and scaffold generation, conformational modeling to capture binding flexibility, and ADMET-aware multi-parametric optimization, along with representative real-world translational case examples. Created in BioRender. Ozdemir, S. (2026) https://BioRender.com/lo6mfgm.

interactions across molecular, cellular, and systemic scales [70]. Many promising leads fail in late-stage development due to poor pharmacokinetics or unexpected toxicities, leading to substantial financial and ethical costs [71]. The integration of generative modeling with ADMET prediction allows these considerations to be incorporated earlier in the design phase rather than evaluated only after candidate selection.

Beyond molecular generation, DL has also made advances in safety-critical tasks across the drug discovery pipeline. For example, recent GNN-based approaches leverage biomedical networks to predict drug-drug interactions for emerging compounds, improving risk assessment in settings where experimental interaction data are sparse and highlighting the expanding role of deep learning in translational pharmacology and drug safety [72].

Generative AI has begun to address these challenges by coupling molecule generation with predictive ADMET modeling. DL models trained on datasets such as ChEMBL [73], protein binding site or protein-protein interfaces [74], small molecules on protein interfaces [75] and large pharmacokinetic registries provide increasingly accurate predictions of solubility, permeability, metabolic liability, and off-target interactions. Generative systems such as ADMET-AI, REINVENT 2.0, and MolDQN explicitly incorporate predicted ADMET properties into the reward or scoring functions guiding molecular sampling [52,76,77]. This enables the simultaneous search for potency, stability, and safety rather than sequential filtering.

Recent DL-based drug discovery studies further illustrate the expanding scope of AI across the pharmaceutical pipeline. For example, Tran et al. introduced MLG2Net, a molecular local-global graph network that integrates chemical structure and pharmacogenomic features to achieve state-of-the-art drug response prediction in lung cancer cell lines, highlighting the growing impact of graph-based deep learning in precision oncology [78].

Recent practical demonstrations show how integrating generative design with ADMET prediction can meaningfully influence candidate quality. One example comes from Insilico Medicine that applied generative reinforcement learning to optimize hepatic clearance and oral bioavailability in early analogs of INS018_055, contributing to sufficient pharmacokinetic stability for advancement to Phase II studies [25]. In another demonstration, Schrödinger and Nimbus Therapeutics used DL-based ADMET prediction coupled with structure-based design to refine TYK2 inhibitors with improved microsomal stability and reduced CYP-mediated metabolism, accelerating lead optimization [79]. These examples illustrate that incorporating ADMET constraints into generative pipelines can reduce late-stage attrition, shift synthesis effort to higher-probability candidates, and allow optimization of efficacy and safety in parallel rather than sequentially.

Across these applications, generative AI shifts drug discovery from passive evaluation to proactive molecular design. Continued progress will depend on coupling generative modeling with automated synthesis, high-throughput experimentation, and iterative feedback loops that reinforce learning from real-world outcomes. Figure 3 summarizes the generative AI applications on drug discovery and their translational cases. These examples demonstrate that generative approaches are not only conceptual tools but have already led to clinically progressing molecules, improved binding selectivity via conformational ensemble modeling, and accelerated property optimization in lead development.

4. Current bottlenecks and emerging solutions in generative molecular design

Generative molecular design has achieved remarkable progress in recent years, yet several fundamental limitations

continue to constrain its translational impact. One major challenge lies in the gap between syntactic validity and true biological or synthetic plausibility. Models based on VAEs, GANs, transformers, and diffusion architectures can readily generate syntactically valid SMILES strings or 3D molecular graphs, but many of these compounds occupy sparsely explored or chemically unrealistic regions of space [80,81]. As a result, extensive post-generation filtering, using synthetic accessibility scores, retrosynthesis planning, and ADMET triage, is typically required to identify plausible candidates. Chemical validity does not equate to biological realism, underscoring the need for stronger integration of synthesis planning, reaction constraints, and prospective experimental validation in generative pipelines [82,83]. In addition to post hoc retrosynthetic filtering, an emerging complementary strategy involves reaction-based generative models, which operate directly on chemical transformations and building blocks rather than atom-by-atom construction. By generating molecules through learned reaction templates or synthesis pathways, these models embed chemical feasibility into the generation process itself, substantially improving synthetic validity and reducing the need for downstream filtering [82,84]. Such approaches offer a promising direction for integrating molecular creativity with practical synthetic constraints.

Beyond these challenges, generative molecular design faces several fundamental limitations and risks that must be addressed for reliable translation. Models are inherently biased toward the statistical distributions of their training data, which can restrict chemical diversity and lead to overrepresentation of well-studied chemotypes while underexploring novel chemical space, novel biological targets and underrepresented protein classes [80,81,85]. Addressing these issues will require broader and cleaner datasets, uncertainty-aware modeling, and integration of multi-modal data sources, such as structural, transcriptomic, and phenotypic information, to improve generalization under out-of-distribution conditions [86,87].

Another limitation is the difficulty of generalizing to novel targets or unseen protein pockets [66]. Many generative models trained on 2D or SMILES representations fail to capture the structural and energetic context of binding, resulting in poor transferability beyond known chemotypes. While recent 3D diffusion and pocket-conditioned frameworks are beginning to address this issue by incorporating spatial interaction constraints, data scarcity and limited availability of high-quality protein-ligand complexes still restrict their generalization potential [88].

Generative frameworks may also produce nonphysical conformers that violate realistic geometric or energetic constraints, particularly when operating without explicit physics-based regularization [89,90]. Moreover, most current models lack direct thermodynamic grounding: they optimize proxy objectives such as docking scores or learned potentials rather than true free-energy landscapes, limiting their ability to capture binding kinetics, solvent effects, and entropic contributions. Reproducibility further remains a critical concern, as stochastic training procedures, incomplete reporting of model architectures and hyperparameters, and sensitivity to random initialization can yield substantially different results across independent runs [82,86,91,92]. Addressing these limitations will require tighter integration of physics-informed modeling, uncertainty quantification, standardized benchmarks, and transparent reporting practices. In addition, public reporting of AI-driven drug discovery remains heavily skewed toward successful or ongoing programs, obscuring negative results and failed campaigns, and further complicating objective assessment of the true causal contribution of generative AI.

In addition to these general challenges, generative molecular design exhibits several systematic and recurring failure modes that limit reliable deployment. RL-guided generation is particularly susceptible to mode collapse and chemotype bias, where optimization converges on narrow regions of chemical space at the expense of structural diversity. In addition, many pipelines over-optimize surrogate objectives such as docking scores or learned binding predictors that correlate imperfectly with true biological activity, resulting in molecules that perform well computationally but fail experimentally. Retrospective benchmark studies may further inflate perceived performance through data leakage, information reuse, or overly permissive evaluation protocols, obscuring real-world generalization. Addressing these failure modes will require robust uncertainty quantification, carefully designed benchmarks, and prospective experimental validation.

Beyond technical limitations, the real-world impact of generative molecular design is strongly shaped by socio-technical and organizational factors that remain insufficiently addressed. Effective deployment requires seamless integration into existing medicinal chemistry workflows and decision processes. Models must be sufficiently interpretable to support expert reasoning, enabling chemists to understand why candidates are proposed and how design trade-offs are navigated. Adoption barriers, including data silos, limited trust in black-box systems, the need for extensive experimental validation, and resistance to workflow disruption, continue to slow translation. In parallel, regulatory considerations impose additional constraints: reproducibility, model traceability, data provenance, and auditability are increasingly critical for regulatory acceptance of AI-supported discovery programs. Addressing these socio-technical challenges will be as essential as continued algorithmic innovation for ensuring durable and responsible impact of generative AI in pharmaceutical development.

RL-guided molecular design introduces another set of challenges. RL models can exploit surrogate reward functions, such as docking or QSAR scores, leading to chemically implausible or mode-collapsed solutions, a phenomenon commonly referred to as reward hacking [93]. Newer approaches emphasize constrained objectives, diversity-aware replay buffers, and hybrid reward shaping to improve stability and prevent over-optimization of proxy metrics [94,95].

Another persistent limitation involves the physical realism of generated molecules. Docking surrogates and deep diffusion models may not accurately capture solvent effects, induced fit, or kinetic processes that govern real binding events [89]. Physics-informed constraints, quantum mechanical priors, and MD refinement should be integrated to ensure that generated molecules correspond to physically meaningful states [90]. Combining these with prospective experimental

benchmarks will be essential for validating generative models beyond *in silico* simulations.

Evaluation and benchmarking present a central bottleneck in assessing generative molecular design [91]. A critical distinction must be made between retrospective validation, which dominates the current literature, and prospective validation, which remains rare but essential for establishing real-world credibility. Retrospective studies, often based on held-out datasets or benchmark libraries, are vulnerable to data leakage, information reuse, and overly permissive evaluation protocols, leading to inflated performance estimates that may not translate to experimental success. In contrast, prospective validation, where model-generated compounds are synthesized and tested in real discovery workflows, provides the only robust measure of practical impact, yet remains limited in scale and scope.

Standardization and reporting practices are therefore crucial. Current studies vary widely in dataset composition, splitting strategies, evaluation metrics, and disclosure of negative results, hindering reproducibility and meaningful cross-model comparison. Minimal reporting standards should include dataset provenance, temporal splitting, full disclosure of training and test separation, hyperparameter sensitivity analysis, and transparent reporting of both successes and failures. Likewise, credible benchmarks should be task-specific and aligned with real discovery objectives, rather than relying solely on generic metrics such as validity, novelty, QED, or synthetic accessibility [86,92]. More constructive alternatives are now emerging. Recent studies advocate the adoption of task-relevant and prospective evaluation metrics, including physics-based stability and energy criteria, pose accuracy under standardized docking protocols, success rates in experimental validation, and performance under prospective blind design challenges [51,82,89,90]. For example, several benchmarks now emphasize molecular stability under energy minimization, consistency across molecular dynamics refinement, and reproducibility of binding poses in independent docking engines. Others evaluate end-to-end performance using prospective synthesis and biological testing rather than static dataset replay. These directions offer a more credible path toward assessing real-world utility than reliance on generic statistical scores alone.

Finally, claims of AI-driven improvement must be contextualized against historical non-AI drug discovery baselines. Historically, conventional drug discovery pipelines required 10-15 years on average from target identification to regulatory approval, with attrition rates exceeding 90% and Phase I-to-approval success rates below 10% [29,71]. Even at the lead optimization stage, only a small fraction of compounds advance from preclinical evaluation into clinical testing, reflecting the inherent uncertainty and complexity of biological systems [71]. While recent AI-driven programs report faster design cycles and encouraging early-phase outcomes, direct comparison remains challenging because these programs often focus on carefully selected targets, benefit from modern experimental infrastructure, and operate under different resource and reporting conditions. Without systematic benchmarking against historical non-AI baselines under comparable constraints, it remains difficult to attribute observed improvements uniquely to generative algorithms rather than to broader organizational, experimental, or strategic factors [51,96]. Likewise, in the absence of direct comparison with traditional medicinal chemistry and physics-based pipelines, apparent performance gains may reflect differences in project selection, resource allocation, or reporting practices rather than true methodological advances. Establishing such baselines is therefore essential for disentangling genuine algorithmic progress from organizational and experimental confounders.

Encouragingly, several research trends are emerging to address these bottlenecks. Diffusion models equipped with SE(3)- or E(3)-equivariant architectures and protein-pocket conditioning are helping to tie molecular generation more closely to realistic interaction patterns [80,88]. Task-relevant and prospective evaluation protocols are being proposed to replace static metrics, while route-aware synthesis planning and retrosynthetic feedback loops are improving the practical feasibility of AI-generated molecules [82,84]. In parallel, RL best practices such as multi-objective balancing and diversity maintenance are enhancing training stability, and uncertainty quantification techniques are helping to manage out-of-distribution risk [94,95].

Despite these methodological advances, a major bottleneck remains the translation gap between computational predictions and real-world validation. While deep generative models have achieved impressive *in silico* results, only a handful of AI-designed compounds or macromolecules have successfully advanced through preclinical development or entered clinical testing [51,81]. This disconnect underscores persistent challenges in modeling complex biological environments, pharmacokinetics, and off-target effects that are not yet reliably captured by current generative pipelines [96]. Bridging this gap will require tighter integration between AI-based design, automated synthesis, and high-throughput biological testing, establishing a closed-loop discovery paradigm in which generative modeling, retrosynthetic planning, and experimental validation are continuously coupled to refine predictions and accelerate translational drug design [51].

5. Conclusion

Generative molecular design has rapidly evolved from a theoretical concept into a practical pillar of computational drug discovery. NN-based architectures have captured complex, nonlinear relationships essential for property optimization; diffusion-based methods have introduced a physics-grounded realism into molecular generation; and LM-based systems have unlocked unprecedented scalability and semantic control over chemical design. Together, these advances have transformed how researchers explore chemical space, moving beyond predefined rules toward adaptive, data-driven discovery.

While major challenges remain, the trajectory of progress is clear. As multimodal datasets, interpretable architectures, and hybrid AI-physics frameworks continue to mature, generative modeling is poised to become a central driver of intelligent, hypothesis-guided molecular innovation. Ultimately, its success will be measured by its ability to accelerate the translation of computational creativity into clinically meaningful therapies.

6. Expert opinion

Over the past few years, we have witnessed an extraordinary acceleration in generative molecular design, driven by the convergence of DL, structural biology, and computational chemistry. Progress has been remarkable, models can now generate chemically valid, diverse, and property-optimized compounds at a scale that was once unimaginable. Together, these advances have shifted the paradigm of computational chemistry from passive screening to active creation. However, this technical success has also fueled the hype, and the field must now confront a central question: which advances truly move drug discovery forward, and which risk becoming elegant but fragile demonstrations?

Still, as with every emerging technology, this progress brings both excitement and caution. Several prevailing narratives deserve careful reexamination. The field has learned, sometimes the hard way, that generating a valid SMILES string or achieving a high docking score does not necessarily mean designing a drug. The real test is not in novelty but in synthesizability, stability, and biological relevance, areas where current systems often fall short. Furthermore, evaluation practices remain inconsistent, with metrics that emphasize form over function. This methodological fragility underscores the need for a stronger connection between computational design and experimental reality.

Yet, the potential of generative AI in drug discovery is immense. If used wisely, it can fundamentally change how we discover, optimize, and validate therapeutic candidates. The ultimate goal should not simply be to generate more molecules but to generate better hypotheses, molecules that are experimentally testable, mechanistically meaningful, and clinically relevant. At the same time, important risks remain underappreciated. Dataset bias and hidden confounders strongly shape model behavior and limit generalization to novel targets and underexplored chemical space. Heavy reliance on surrogate objectives, such as docking scores or learned binding predictors, encourages reward hacking and over-optimization of metrics only weakly correlated with biological success. Benchmark inflation and inconsistent reporting further obscure failure modes and inflate perceived progress. Moreover, causal attribution of recent successes remains unresolved: improvements often reflect advances in experimental infrastructure, target selection, and organizational strategy as much as algorithmic innovation.

Some directions may also represent likely dead ends. Purely in silico pipelines without tight experimental feedback are unlikely to achieve robust translational success. Scaling model size without corresponding improvements in data quality, biological annotation, and evaluation rigor risks diminishing returns. Overemphasis on static affinity prediction, without accounting for kinetics, dynamics, and cellular context, will continue to constrain clinical relevance.

In light of these realities, the next phase of generative molecular design should prioritize four interdependent challenges. First, deep integration with experimental platforms is essential to close the design-make-test loop and provide continuous feedback. Second, richer multimodal datasets capturing chemical, structural, and biological context are required to improve generalization and mechanistic grounding. Third, evaluation must shift from retrospective benchmarks toward standardized, task-relevant, and prospective validation. Finally, physical and biological realism, through physics-informed constraints, uncertainty quantification, and mechanistic modeling, must become core design principles rather than afterthoughts.

Encouragingly, several developments already align with these priorities. Structure-aware diffusion models conditioned on protein pockets now learn realistic spatial interaction patterns, enabling ligand generation in 3D while accounting for steric fit, hydrogen bonding, and conformational adaptability. The recent advances [97] in structural coverage of human proteome and interactome [98] have expanded our horizon. The expanding availability of high-quality predicted protein structures has further opened generative design to targets previously considered inaccessible [45,46]. Energy-aware modeling over Boltzmann-distributed conformational ensembles is beginning to move the field beyond static affinity optimization toward more realistic binding dynamics [47]. Reinforcement learning frameworks that incorporate synthetic accessibility and ADMET constraints increasingly produce decision-ready, experimentally viable candidates.

Looking forward, the convergence of generative AI with automated synthesis planning is poised to fundamentally reshape medicinal chemistry and pharmaceutical development. By embedding route-aware retrosynthesis, feasibility scoring, and reaction prediction directly into generative design pipelines, researchers can prioritize molecules that are not only potent and selective but also synthetically tractable at scale. This integration enables continuous design-make-test cycles in which computational models propose candidates, synthesis platforms evaluate and execute feasible routes, experimental data are rapidly generated, and model parameters are iteratively refined. The practical impact of this closed-loop paradigm is profound: medicinal chemists can explore broader chemical space with higher confidence, reduce costly late-stage failures, compress development timelines, and shift effort toward candidates with the highest probability of translational success. As these technologies mature, generative AI will move from a supportive tool to a central organizing framework for modern drug discovery and development.

Looking ahead, we expect the field to move decisively toward foundation models trained across multiple biochemical modalities, models that can reason simultaneously about chemistry, structure, and biology. These systems will not just generate molecules but infer how they interact with proteins, cells, and pathways. In the near term, hybrid workflows, including generative models proposing candidates, physics-based simulations refining them, and wet-lab validation closing the loop, will likely dominate. Over time, this integration will blur the boundaries between computational prediction and experimental discovery.

Alternative approaches, such as quantum-enhanced modeling, active learning with robotic synthesis, and large-scale graph neural frameworks, will complement generative AI by introducing new layers of physical fidelity and automation. The key will be to build pipelines that integrate these technologies seamlessly rather than treating them as competitors.

The central challenge will not be the invention of additional model classes, but the disciplined integration of these technologies into coherent, experimentally grounded pipelines.

In our view, generative modeling is no longer a speculative technology. It is becoming the scientific engine of modern drug discovery. The next decade will determine how effectively we can translate its creative potential into therapeutic impact. If the community succeeds in integrating data, physics, experimental feedback, and the willingness to confront uncomfortable limitations within a unified framework, generative AI will transform the field from a search through chemical space into a guided, intelligent exploration of biological possibility.

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Declaration of interest

The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed. The authors declare no competing interests.

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